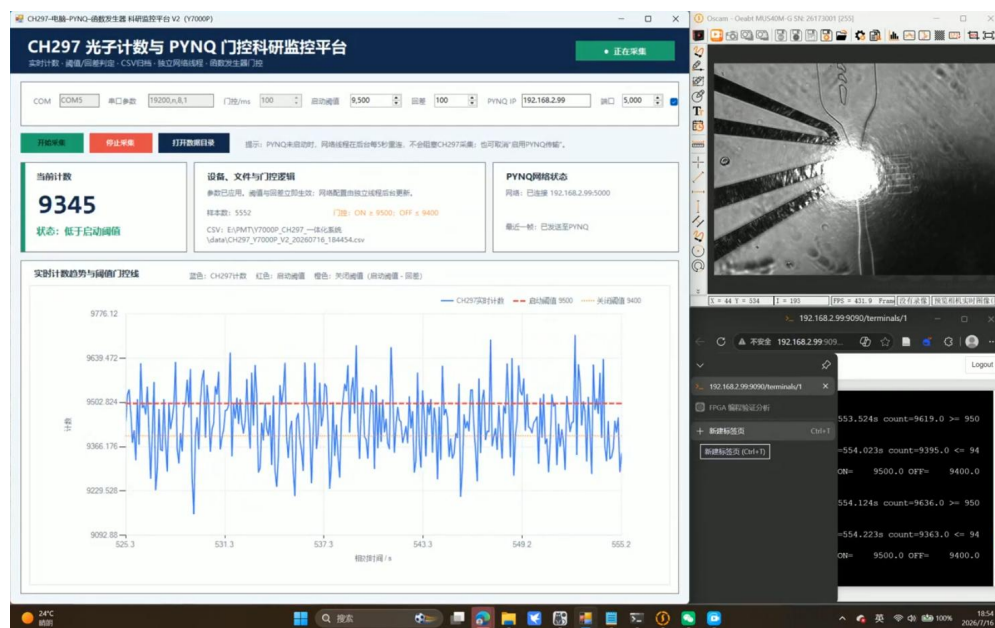


Lab Notebook (Microfluidic Platform)

August

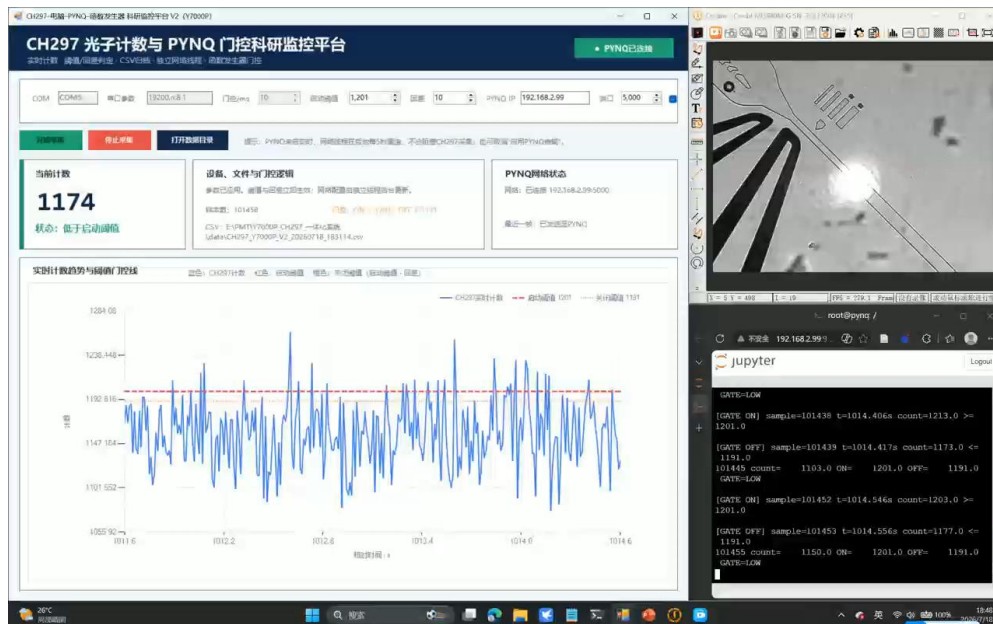
Dielectrophoretic (DEP) Sorting Function

We chose Solution 2 and independently developed the first-generation control software to unify control of the entire microfluidic platform; when testing the sorting function, we found that oversized droplets could not be sorted.



Video 9 — Initial DEP sorting attempt

Subsequently, smaller droplets were generated and the laser-detection gate was narrowed, successfully obtaining the result that fluorescence-excitabile droplets were sorted by dielectrophoresis.

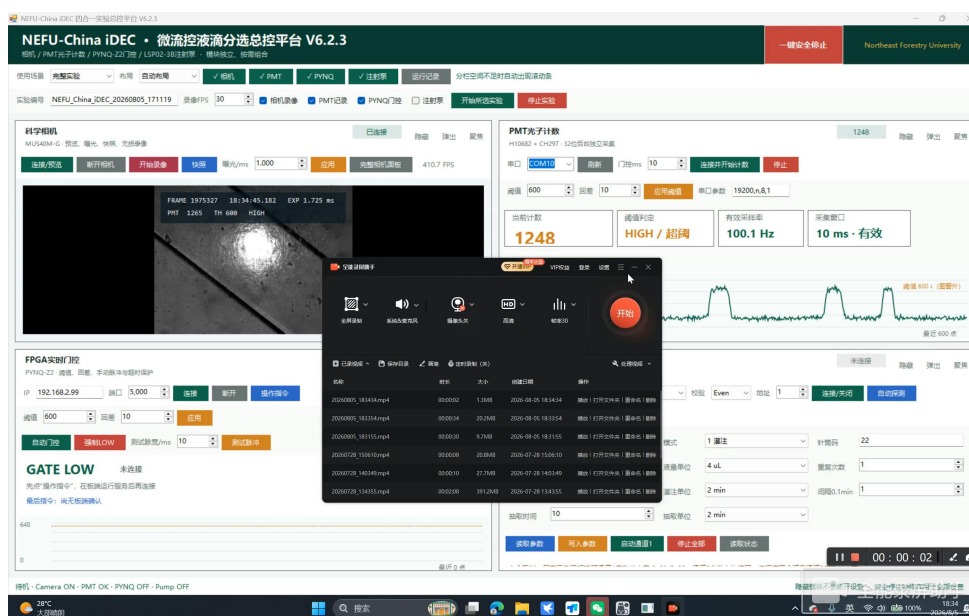


Video 10 — Successful DEP sorting of fluorescent droplets

Fluorescence-Discrimination Validation

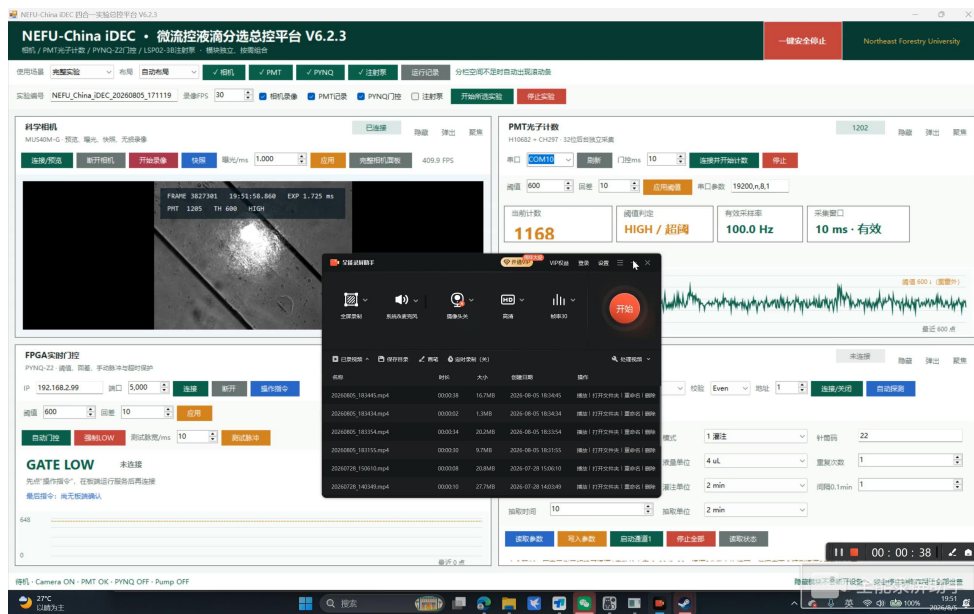
Fluorescence photon counts:

induced ≈ 1000 (2250–1250);

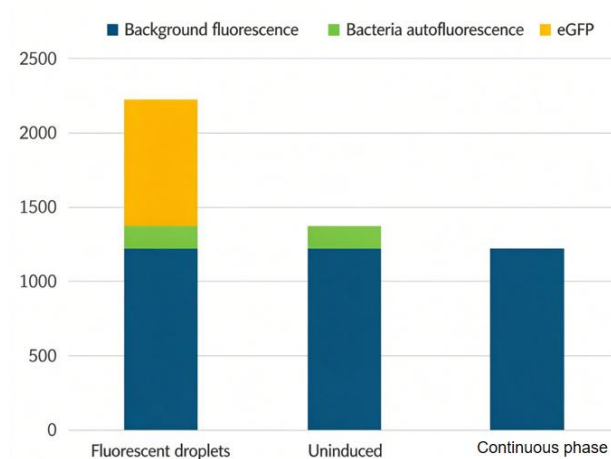


Video 11 — Induced droplet fluorescence

uninduced ≈ 150 (1350–1200); background ≈ 1225 .

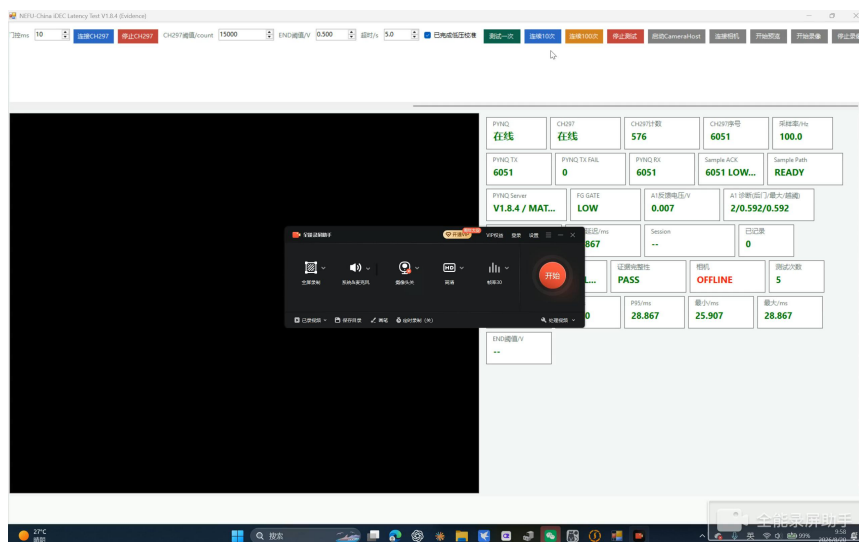


ideo 12 — Uninduced droplet fluorescence



System Latency Test

The interval from the LED's momentary flash to the system's control signal was recorded, yielding an average latency of 25.359 ms (22.785 + 2.574 ms).



Video 13 — System latency test

Calculation method	Corresponding time	Converted droplet rate
Fastest successful sample loading	18.735 ms	53.4 per second
Average sample loading	25.359 ms	39.4 per second
Average + 1 standard deviation	28.432 ms	35.2 per second
Actual measured successful sample loading	30.923 ms	32.3 per second
Average + 2 standard deviations	31.505 ms	31.7 per second

Time period	Average value (ms)	Standard deviation (ms)	Minimum value (ms)	Maximum value (ms)
Fastest	28.735	22.95	0.010	28.125
Average	28.439	25.4	0.121	255.2
最快流动速率	28.432	25.55	8.07s	14.735

Experimental results / 100 trials
Average: 22.785 ms + 2.574 ms = 25.359 ms; main result is 25.359 ms. The comparison does not involve dead time and hysteresis, so only the average value is considered. Experimental result A1 did not reach 0.5 V.

All videos referenced in this document are available for download at the bottom of our team's Notebook page.